

PAPER394 — F</i>U Complete Three-Term Star Magic Master Equation

Session: 107 (groksharecfdcad2f5.txt deep re-analysis pass)

PAPER394 — FU Complete Three-Term Star Magic Master Equation

Source: groksharecfdcad2f5.txt, lines ~200–1600 (C++ compute_FU() + simulation outputs) **Section:** main.cpp compute_FU() function + "Program Execution Summary" Grok response **Session:** 107 (groksharecfdcad2f5.txt deep re-analysis pass) **CP4 Class:** FUThreeTermStarMagicMasterCalculator (CP4 #45)

1. Overview

PAPER326 introduced the **Triadic Master Equation** for F_U with three co-equal Aether arms. PAPER394 upgrades this to the **complete four-Ug plus buoyancy plus magnetic plus metric tensor** form that is actually implemented in the Grok simulation, with confirmed numerical outputs for all four test bodies.

The complete implementation reveals:

Four Ug gravitational-field terms (not three)

Explicit buoyancy sum $\sum U_{bi}$

Magnetic string term U_m

Aether metric tensor trace $\text{tr}(A_{\mu\nu})$

k-constants: $k_1=1.5$, $k_2=1.2$, $k_3=1.8$, $k_4=2.0$

2. The Complete F_U Formula

2.1 Master Equation (Full Form)

$$F_U = \sum_{i=1}^4 (U_{gi} + U_{bi}) + U_m + \text{tr}(A_{\mu\nu})$$

2.2 Expanded Form

$$F_U = (U_{g1} + U_{b1}) + (U_{g2} + U_{b2}) + (U_{g3} + U_{b3}) + (U_{g4} + U_{b4}) + U_m + \text{tr}(A_{\mu\nu})$$

2.3 Individual Term Definitions

Term 1 — Magnetic Dipole (Ug1):

$$U_{g1} = k_1 \cdot \mu_s(t) \cdot \nabla M_s/r \cdot e^{-\alpha t} \cdot \cos(\pi t_n) \cdot \delta_{\text{def}}(t)$$

$$k_1 = 1.5$$

Term 2 — Charge-Reactivity (Ug2):

$$U_{g2} = k_2 \cdot (Q_A + Q_{UA}) \cdot \frac{M_s}{r^2} \cdot S(r, R_b) \cdot w_{\text{sw}} \cdot H_{\text{Scm}} \cdot E_{\text{react}}$$

$k_2 = 1.2$

Term 3 — String Rotation (Ug3):

$$U_{g3} = k_3 \cdot B_j(t) \cdot \cos(\omega_s(t) \cdot \pi \cdot t) \cdot P_{\text{core}} \cdot E_{\text{react}}$$

$k_3 = 1.8$

Term 4 — Vacuum Concentration / BH Coupling (Ug4):

$$U_{g4} = k_4 \cdot \rho_v \cdot C_{\text{conc}} \cdot \frac{M_{bh}}{d_g} \cdot e^{-\alpha t} \cdot \cos(\pi \cdot t_n) \cdot (1 + f_{fb})$$

$k_4 = 2.0$

Buoyancy (Ubi for each Ugi):

$$U_{bi,i} = -\beta_i \cdot U_{g,i} \cdot \omega_g \cdot \frac{M_{bh}}{d_g} \cdot (1 + \epsilon_{sw} \cdot \rho_{sw}) \cdot [UA] \cdot \cos(\pi \cdot t_n)$$

Magnetic Strings (Um):

$$U_m = \frac{\mu_j(t)}{r_j} \cdot (1 - e^{-\gamma t} \cos(\pi \cdot t_n)) \cdot \hat{\Phi} \cdot n_{\text{strings}} \cdot P_{\text{SCm}} \cdot E_{\text{react}}$$

Aether Metric Tensor Term:

$$\text{tr}(A_{\mu\nu}) = \text{tr}(g_{\mu\nu}) + 4\eta \cdot T_{s00} \cos(\pi \cdot t_n) = -2 + 4\eta \cdot T_{s00} \cos(\pi \cdot t_n)$$

3. k-Constants Table

Constant	Value	Term	Physical Role
k_1	1.5	Ug1	Dipole magnetic coupling weight
k_2	1.2	Ug2	Charge-reactivity coupling weight
k_3	1.8	Ug3	String rotation coupling weight
k_4	2.0	Ug4	Vacuum-BH concentration weight
\beta_i	0.6	Ubi	Buoyancy ratio coefficient

4. Simulation Verification

4.1 FU Outputs at t=0, r=R_b

Body	R_b (m)	M_s (kg)	FU (normalized units)
Sun	1.496×10^{13}	1.989×10^{30}	$-2.063905868374393 \times 10^{59}$
Earth	1×10^7	5.972×10^{24}	$-2.0639058683743924 \times 10^{53}$
Jupiter	1×10^8	1.898×10^{27}	$-2.0639058683743924 \times 10^{54}$
Neptune	5×10^7	1.024×10^{26}	$-2.0639058683743926 \times 10^{52}$

4.2 Dominant Term Analysis

The output $F_U \sim -10^{59}$ for the Sun is driven by:

$$U_{g3}(\text{Sun}) \sim k_3 \times B_j \times E_{\text{react}} = 1.8 \times 10^3 \times 8.808 \times 10^{54} \approx 10^{58}$$
$$U_{bi3}(\text{Sun}) \approx -0.6 \times U_{g3} \rightarrow \text{also } \sim 10^{58}, \text{ negative}$$

The net negative sign of F_U arises because $\Omega_g = 7.3 \times 10^{-16}$ and $U_{bi,i}$ terms slightly dominate due to the M_{bh}/d_g coupling, making the buoyancy terms collectively exceed the gravitational terms.

4.3 Structural Pattern

The ratio between bodies scales with their characteristic radii:

$$F_U(\text{Sun})/F_U(\text{Earth}) \approx 10^{59}/10^{53} = 10^6$$

For $R_b(\text{Sun})=1.496 \times 10^{13}$ vs $R_b(\text{Earth})=10^7$: ratio = $1.496 \times 10^6 \approx 10^6 \checkmark$

5. Comparison to Existing Papers

Paper	F_U Form	What's Missing
PAPER_326	3-arm Triadic	No Ug4, no explicit Ubi sum, no metric term
PAPER_345	Partial expansion	Missing k-constants verification
PAPER_350	vSCm in Ereact	No complete combination
PAPER_394	All 4 Ug + Ubi + Um + tr(A)	Complete form + verified outputs

6. Global Parameters

Parameter	Value	Role
Ω_g	7.3×10^{-16} rad/s	Galactic angular frequency
M_{bh}	8.15×10^{36} kg	Central BH mass (SgrA*: $4.1M_{\odot}$)
d_g	2.55×10^{20} m	Distance to galactic center
ρ_v	6×10^{-27} kg/m ³	Vacuum energy density
C_{conc}	1.0	Concentration factor
n_{strings}	10^9	Number of magnetic strings

7. Summary

PAPER_394 documents the **complete implementation** of the UQFF unified field strength F_U as used in the Grok simulation. The formula contains four Ug terms ($k_1=1.5, k_2=1.2, k_3=1.8, k_4=2.0$), four corresponding buoyancy corrections, magnetic string term U_m , and Aether metric tensor trace $\text{tr}(A_{\mu\nu})$. Verified simulation outputs confirm $F_U(\text{Sun}) = -2.064 \times 10^{59}$ with Ug3 as the

dominant driving term via $E_{\text{react}} = 8.808 \times 10^{54} \text{ J}$.